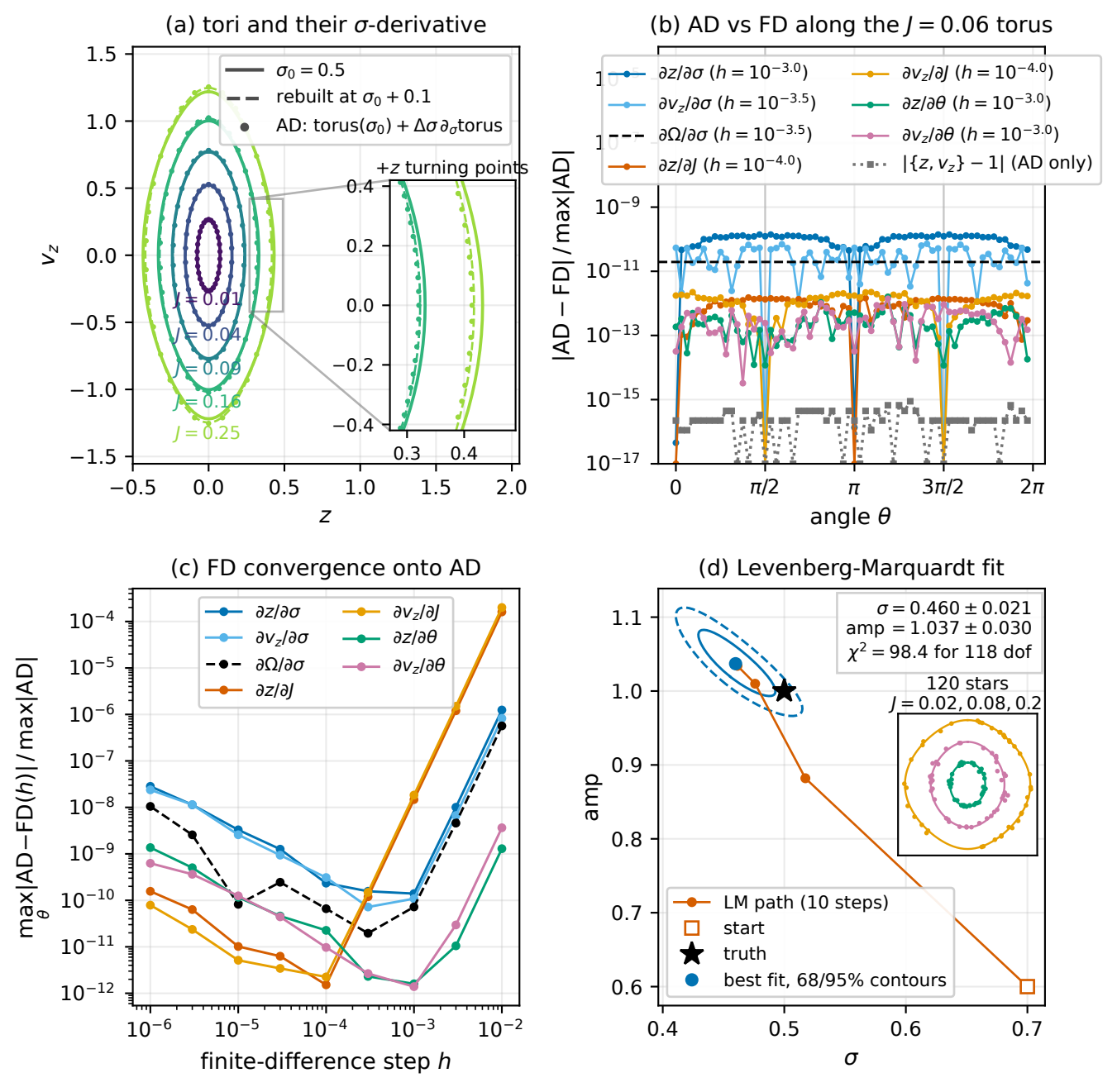

(a) tori and their σ-derivative
σ0 = 0.5
rebuilt at σ0 + 0.1
AD: torus(σ0) + Δσ ∂σtorus
+z turning points
J = 0.01
J = 0.04
J = 0.09
J = 0.16
J = 0.25
vz
z
(b) AD vs FD along the J = 0.06 torus
∂z/∂σ (h = 10−3.0)
∂vz/∂σ (h = 10−3.5)
∂Ω/∂σ (h = 10−3.5)
∂z/∂J (h = 10−4.0)
∂vz/∂J (h = 10−4.0)
∂z/∂θ (h = 10−3.0)
∂vz/∂θ (h = 10−3.0)
|{z, vz} − 1| (AD only)
|AD − FD| / max|AD|
angle θ
(c) FD convergence onto AD
∂z/∂σ
∂vz/∂σ
∂Ω/∂σ
∂z/∂J
∂vz/∂J
∂z/∂θ
∂vz/∂θ
max|AD−FD(h)| / max|AD|
finite-difference step h
(d) Levenberg-Marquardt fit
σ = 0.460 ± 0.021
amp = 1.037 ± 0.030
χ2 = 98.4 for 118 dof
120 stars
J = 0.02, 0.08, 0.2
LM path (10 steps)
start
truth
best fit, 68/95% contours
amp
σ